Behind the Seams

Philippine Textiles Research Institute
Annual Report 2018

ABOUT

About this Report

As the premier textile research and development arm of the Department of Science and Technology (DOST), the Philippine Textile Research Institute (PTRI) has long been at the forefront in addressing the new challenges and opportunities facing the Philippine textile industry. The 2018 Annual Report proudly showcases the biggest milestones and achievements of the Institute for the year. The report also looks into what the PTRI has in store for the succeeding year as it continues to develop Philippine weaving craft and to offer technical services to micro-, small, and medium enterprises (MSMEs) in the industry, thus contributing to nation-building.

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OUR INDUSTRY AT A GLANCE

Interest in local products and Filipino fashion has resulted in a resurgence of traditional crafts and has encouraged the modernization of Filipino weaving and textiles.

The Philippine Textile Research Institute (PTRI) is committed to upholding the highest standards of excellence in all of its outputs and services. The Institute remains a steadfast partner of large companies and particularly micro-, small, and medium enterprises (MSMEs) in the development, testing, and quality assurance of fibers/yarns and other raw materials. It also provides skills enhancement to improve productivity among workers in the textiles and garments industry.

In 2018, textile manufacturers contributed 5.7% (Php 5,840 million) of the growth in the manufacturing sector. Thanks in part to PTRI's efforts, the Philippine textiles industry remains a top employer, accounting for more than 20% of employment in the manufacturing sector. While industry players have downsized in recent years, the textiles industry is still seen as having immense potential in generating much-needed jobs in the country.

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Our Value Chain

Mandated by Executive Order No. 128, the PTRI dedicates itself to the study, preservation, and development of Philippine textiles.

[Infographic 1] PTRI Mandate

- Conduct applied research and development for the textile industry sector;
- Transfer the completed research to end-users or via linkage units of other government agencies; and
- Undertake technical services and provide training programs for various stakeholders.

[End of infographic1]

To start with, the Institute conducts a careful research of the needs and expectations of the industry stakeholders. Research/project proposals from the technical groups are prepared and submitted for review by the Institute's Executive Committee, with inputs from the Planning and Finance & Administrative Groups. From there, these proposals undergo assessment in terms of resource availability; risk management/management review; monitoring, analysis, and evaluation; product conformity; and customer satisfaction. Once approved, each project goes into implementation, pilot testing, packaging, and finally rollout or commercialization.

[Infographic 2] PTRI Business Process Map
(Insert graphic from page 16 of PTRI Annual Report 2017)
[End of Infographic 2]

In pursuing its mandate, the PTRI keeps apace with industry developments and responds to the various challenges and opportunities presented in the domestic and

international markets. The Institute promotes sustainable practices among textile manufacturers and creates solutions that respond to the demand for contemporary and eco-friendly alternatives to existing products.

[Infographic 3] PTRI's Industry Interventions in 2018

Fiber production
Stakeholders: Fiber crop producers, sericulturists, cooperatives, etc.
- Introduction of silkworms to produce commercially viable natural fibers

Spinning
Stakeholders: Yarn manufacturers, multi-purpose cooperatives, etc.
- Development of indigenous fiber-based yarns

Fabric production
Stakeholders: Fabric manufacturers, weavers, etc.
- Development of smart textiles

Dyeing and finishing
Stakeholders: Textile dyeing and finishing mills
- Color profiling of natural dyes and other technical support

Clothing production
Stakeholders: Garments manufacturers
- R&D and prototyping of flame-retardant and mosquito-repellent clothing
- Textile-related assistance resulting in eco-friendly products for MSMEs

Clothing retailing
Stakeholders: Clothing retailers
- Design competitions and support for MSME participation in industry trade fairs

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The diverse fields of Filipino culture, science, art, and business converge at the heart of the Philippine textile industry. PTRI recognizes that collaboration, through a combination of expertise among the various sectors, is integral to developing the entire industry.

The Institute places great value in inclusiveness and assists the country's various artisans, weavers, and farmers to become more globally competitive. In this way, the Institute also contributes to nation-building.

Through the work and expertise of the Institute, different sectors within the industry are able to strengthen their competencies and bring their distinctly Filipino products into the consciousness of consumers.

Messages

Message from the Secretary

[Waiting on answers to questionnaire]

Message from the Director

[Waiting on answers to questionnaire]

ABOUT PTRI

The PTRI is strengthened by the core values and principles that the Institute and its staff aim to exhibit in all of their work. Guided by its mission and vision, the Institute commits itself to achieving excellence and ground-breaking developments in the textile industry for the benefit of the Filipino people.

Vision

To be the Center of Excellence on textile science, technology, and innovation for sustainable and globally competitive Philippine textile, garment, and allied industries.

Mission

To support the Philippine textile, garment, and allied micro-, small and medium enterprises (MSMEs), and achieve global competitiveness with unwavering commitment to work for excellence and integrity through the:

- Judicious utilization of textile-based resources;
- Development and transfer of innovative technologies;
- Enhancement of human resource capabilities; and
- Promotion of quality assurance and quality control,

all of which, with unwavering commitment to excellence and integrity.

Our Core Values

We are here to deliver our products and services in a manner that satisfies our customers. Customer service entails professionalism, technical competence, reliability, and integrity in order to provide excellent service at all times.

Professionalism

The high-caliber research and technologies developed by the PTRI is a testament to the level of professionalism that its personnel and leadership possess. The PTRI serves as a hub of innovative technologies. Moreover, collaborations with designers, business and other industry partners offer opportunities for exploration, iteration, and further development of PTRI-led technologies.

Technical Competence

We provide our services based on our competence and knowledge of the garment and textile industry. We encourage the PTRI staff to explore continuing education opportunities to expand their capabilities. We are committed to providing guidance and mentoring toward the pursuit of excellence.

Reliability

We support our stakeholders through advanced research, laboratory services, and government as well as non-government linkages. We ensure that our products and services satisfy our customers. Quality is assured given that PTRI is continuously improving its methodologies through PNS ISO/IEC 17025:2005 (Chemical and Mechanical Testing)

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accreditation and ISO 9001:2015 (Quality Management System) certification. As a reliable partner of the garment and textile industry, we maintain our reputation at all times.

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Integrity

We adhere to the high ethical principles and professional standards for public officials and employees. We strive to be trustworthy and transparent in all our dealings.

OUR PERFORMANCE

Concept to Cloth - Translating Ideas to Reality

Philippine textiles are cultural artifacts that represent the natural ties and indigenous traditions of Filipinos. The PTRI believes in the potential of textiles to increase in value as commodities. They will not only promote an appreciation of Filipino heritage, but also become instrumental in improving the livelihood of Filipinos and uplifting the poorest local communities.

In all of its decades of service, the PTRI has remained committed to its vision of developing a robust, globally competitive textiles industry, backed by its flagship campaign of "Textiles Empowering Lives Anew (TELA)." The latter serves as a rallying cry, and combines resource management, knowledge-sharing, technology transfer, and promotion of the unique qualities of Philippine textiles and textile-based products.

In its 51st year, the Institute proudly launched "Concept to Cloth," which is focused on translating ideas into reality. The 2018 theme homes in on the combined efforts of the government, the academe, and the private sector in generating ideas, and turning those ideas into reality. This involves conceptualizing all of the different possibilities for advancing Philippine textiles, and following through with the creation of programs, products, and systemic solutions to further improve the textile industry.

This report faithfully documents the various ways by which PTRI and its allies have followed through on this promise.

[Infographic 4] **Accomplishments for Fiscal Year 2018**

TEXTILE AND OTHER TEXTILE-RELATED R&D PROGRAM

143%
Percentage benefit incidence of intervention to local industries and/or institutions

35
(above target of 2)
No. of partnerships with public and private stakeholders and international organizations

11
(above target of 9)
No. of completed projects for the period

100%
(on target)
Percentage of projects implemented within the approved timeframe

14%
(on target)
Percentage of projects completed which were published in peer-reviewed journals, presented in national and/or international conferences, or with IP filed or approved

TEXTILE S&T SERVICES PROGRAM

97.53%
(above target of 92%)
Percentage of customers that rate the technical services rendered as satisfactory or better

22,552
(1,357 in excess of target)
No. of technical services rendered

100%
(above target of 95%)
Percentage of request for technical services that have been provided within the required timeframe

Number of clients benefiting from technical services

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TEXTILE TECHNOLOGY TRANSFER PROGRAM

94.23%
(above target of 90%)
Percentage of clients that rate the technology transfer as satisfactory or better

92%
(above target of 90%)
Percentage benefit incidence of intervention to local industries and/or institutions

60
(above target of 58)
No. of knowledge/technologies diffused

[End of Infographic 4]

OUR DIVISION HIGHLIGHTS

[Division 1] **Research and Development**

This 2018, the PTRI achieved breakthroughs at every step of the textile production process, starting with its primary function of research and development (R&D). Continuous R&D is essential to refining the knowledge within the industry, developing facilities and training modules for better performance and efficiency, innovating on fabrication, and strengthening cooperation through partnerships with different institutions, businesses, and communities.

Sericulture

SILK FROM LOCAL SILKWORMS
Revitalization of silk industry in northern Mindanao
(Phase III)

In the past 40 years, the PTRI's Sericulture Technology Center in Misamis Oriental became a reliable source of high-hatching, disease-free silkworm hybrid eggs. The Center established four patented Mindanao silkworm hybrids, an achievement that ended the need for silkworm egg importation and elevated Misamis Oriental as a leading province in sericulture in the country.

Despite this feat, however, the sericulture industry in southern Philippines did not progress due to continuing challenges: an unstable market, high production costs, low technology adoption by farmers, better-priced imported alternatives compared to locally produced cocoons and raw silk, lack of participation from the private sector, and lack of silk yarn processing facilities.

The PTRI therefore aimed to revitalize the silk industry in Northern Mindanao by developing fresh hybrid silkworm lines and, moving beyond cocoon production technology, equipping the center with the capability to produce commercial silk, which will then be used to manufacture silk yarns, fabrics, and other products for the local and international markets.

The Institute kicked off the three-year project by identifying quality silkworm parent breeds, originating from Benguet, which could thrive in the conditions of Misamis Oriental. By the end of the project, the PTRI successfully established eight new silkworm hybrids, which served as back-up to the existing Mindanao hybrids.

In 2018, the Center was able to produce 720kg fresh cocoons which translated into 72kg of raw silk. Three sets of raw silk were made into shawls and sarongs that were colored with locally available natural dyes. Meanwhile, five sets of raw silk were twisted to produce thrown silk threads, ready for use in the production of other handwoven products.

The eight new hybrids has genetically high hatching rates that could further be enhanced through proper rearing practices. While the PTRI originally intended to study methods on increasing the productivity of the silkworms in a one hectare field, due to the late planting of the mulberry trees this was not able to push through. However, these plans are expected to be in full swing by 2019.

PTRI ESTABLISHES SILKWORM GENETIC DATABASE
Genome-based approach silkworm germplasm maintenance and hybridization (Phase I)

As part of the Philippine Silk Road Program which aimed to increase the country's capabilities for silk yarn production, the Institute conducted a study on the genes of *Bombyx mori,* or the domesticated silkworm, in order to classify the different silkworm gene strains that are available in the country. From 17 pure parental lines, the study was able to identify 13 well-performing strains, and four poor-performing strains. The identification of these strains were crucial in marker assisted selection (MAS) and in the mapping of quantitative trait loci (QTL), which would allow them to select preferred traits of the silkworms during the breeding process.

The silkworm samples were collected from the PTRI Technology Center in Misamis Oriental (TCMO), DOST-Cordillera Administrative Region (DOST-CAR), and Sericulture Research and Development Institute (SRDI). Before the collection of the tissue samples, the silkworms underwent phenotypic and morphological characterization from egg to larvae. The samples then underwent genomic DNA extraction to assess their DNA integrity, COI amplification, purification and sequencing, *in silico* assessment, and data mining.

By the end of the study, the Institute was able to successfully establish (1) a protocol for the extraction of good-quality DNA from live silkworm samples for accurate genetic characterization and (2) create a genomic database of the gene strains. The database—including DNA barcodes, genotyping by sequencing (GBS) library, metagenomics library containing genetic material recovered directly from environmental samples—will be used in future marker assisted breeding activities and will serve as a reference in comparing the traits to international standards. This will ultimately redound to the availability of superior hybrids for the sericulture industry.

SUSTAINABLE SILKWORM REARING
Revitalization of Sericulture in Cordillera Administrative Region (CAR)

Following the decline of the local sericulture industry in the year 2000, the PTRI, in collaboration with the DOST-CAR, has placed a tremendous amount of effort in revitalizing the sericulture industry in the Cordillera Administrative Region. Encouraged by the presence of potential weaving partners (Sagada Weaving and Narda's), and sericulture partners (Kalinga State University and Apayao State College), the project aimed to strengthen the network and cooperation of the various stakeholders and institutions working for the development of sericulture and maintenance of germplasm. This is in the hopes of creating a stable source of quality silkworm eggs, and a sustainable stream of production.

As their first step towards capability-building, two new staff members were sent to undergo germplasm management and rearing training at DOST-CAR. One of the staff members was even able to attend the "Mulberries' Sericulture Training" in Laos where participants were able to learn about the sericulture production cycle, including topics on silkworm rearing, reeling, natural dyeing, and weaving.

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Maintenance rearing was conducted on the existing 18 mainlines and two parent lines in the facility. The activities were conducted in the existing area provided by the DOST for sericulture. Aside from maintaining 18 silkworm purelines, the facility also supplies hybrid eggs to farmers in Kapangan, Kibingan, Bakun in Benguet, and in Sabangan, Mountain Province.

Meanwhile, portions of land owned by the Kalinga State University and the Apayao State College were utilized for mulberry cultivation. At the start of the rainy season, they were able to plant a total of 40,000 mulberry cuttings in 24 plots for rooting and immediate transplanting. KSU was also able to conduct a test rearing of 1/5 box of silkworm eggs, producing 5.2kg fresh cocoons.

Lastly, the project established a brand new filature and was able to procure a cold storage facility and a boiler ready for installment. They were able to finish the procurement of filature equipment, which are set to be installed in the first quarter of 2019.

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Natural and indigenous fibers

THE NEXT MARKETABLE GREEN FIBER
Pilot scale verification/validation of the pre-treatment of bamboo fiber

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You may opt to use just the formal project titles below.

The PTRI has always given great importance to the development of locally sourced raw materials that may benefit the textile industry. The recent trend has been to develop indigenous natural fibers—this includes silk, piña, banana, abaca, maguey, water hyacinth/lily, saluyot, sisal, salago, and buri—for textile applications that could compete in the international market.

Natural fibers are being used for the automotive, medical, sports/leisure industry, among others. Due to their biodegradable and renewable qualities, natural fibers are now widely considered as eco-friendly and cost-efficient alternatives for synthetic fibers. Moreover, the use of natural fibers presents an opportunity to use agricultural byproducts for textiles, thus providing additional livelihood and income for the farming community.

Bamboo fiber in particular has been a material of interest as a viable natural fiber. World trade in bamboo and bamboo products has been growing at an exponential rate since 2002. Besides being biodegradable, bamboo fiber is naturally antibacterial, UV protective, breathable, durable, and graced with a luxurious patina.

Also referred to as “green fiber,” bamboo can be selectively harvested annually, and naturally regenerates without replanting. Compared to cotton, the grass requires less water to maintain and has no negative environmental impacts. In fact, bamboo is known to improve watersheds, purify air quality, remove toxins from contaminated soil, and help prevent soil erosion—it can absorb up to 12 tons of carbon dioxide per hectare and it produces 30% more oxygen than any hardwood forest of similar size.

With this in mind, the PTRI set out to determine the commercial potential of bamboo in being used as a natural fiber for textiles and fabrics.

After subjecting 26 batches (192.9kg) to chemical-mechanical treatments to extract the fiber, it was found that bamboo had a higher fiber recovery rate—meaning less of the material is lost during processing—compared to abaca, banana, pineapple, saluyot, and water hyacinth.

The treated bamboo fiber was soft and pliable and could be blended with polyester and cotton fiber. Comparatively, the resulting polyester-blended bamboo fiber had a finer yarn count, a lower tendency for breakages compared to cotton-blended bamboo fiber, and relatively higher overall fiber recovery. While the tested procedures were conducted in the lab, the proponents of the study confirmed that the process can successfully be upscaled for commercial production.

THE BEST SPINNING METHOD FOR NATURAL FIBERS
Development of indigenous fiber-based yarns with higher bends using worsted spinning systems

The PTRI has long been at the forefront in developing yarns made from natural fibers blended with cotton and/or polyester staples using the short staple system—a fiber spinning method meant to produce highly compact yarns that have fewer protruding fibers (aka hairiness), better evenness, and fewer imperfections. However, the method still faces limitations in production and the quality of the output. It should be noted that the resultant yarn composition still contains low indigenous fiber content (for example, 80% cotton or polyester blended with 20% indigenous fiber). Other undesired qualities include coarseness and unevenness in the yarn and, subsequently, a high degree of fabric pilling (loose strands or balls that form on a piece of fabric).

The study set out to determine which spinning system was best for processing pineapple, abaca, and banana blended yarns. Separate batches of pineapple, abaca and banana fibers underwent pretreatment as preparation for the study. Each batch was then blended with polyester with a ratio of 60/40, 50/40, and 40/60 polyester/natural.

All the batches underwent processing through the semi-worsted spinning system, which was deemed suitable for processing natural fiber as it requires just the right staple feed length compared to the cotton and worsted spinning systems. (Natural breakages during the spinning process were observed to determine which method produced more even and much stronger yarns and, consequently, “less itchy” fabrics.) The spun yarns were woven into cloth by the Weaver Textile Mills Inc.

Results of the study showed the semi-worsted system increased the textile fiber composition of yarns to 60% and is comparable in terms of breakages per spindle rotation to a similar yarn composition produced using the cotton system.

BETTER TREATMENT FOR NATURAL FIBERS
Modified pretreatment of indigenous fibers for textile application

An increased awareness of environmental and social issues by the public has created a demand for the integration of sustainable and eco-friendly textiles in the local and international markets. With the growing interest spurred by these trends, the PTRI conducted studies on the commercial potential of natural fibers, which has led to the use of pineapple, abaca, water hyacinth and banana fibers in textile production.

While the use of natural fibers has been promising, the process is not without its challenges. One problem in particular is the limited amount of natural fiber that can be combined with either cotton or polyester, and the presence of protrusions along the yarn which results in a fabric that is uneven and itchy in texture. The Institute thus set out to improve existing pretreatment methods in order to enhance the overall physical-mechanical properties of indigenous fiber blended yarns.

Using dilute mineral acid during acid prehydrolysis, the Institute was able to enhance the penetration of degumming chemicals into the microfibril structure of abaca and pineapple fiber resulting in better removal of non-cellulosic components like lignin and hemicellulose. The method improved the spinning performance of the fiber, lowered the tendency of breakages (ends-down), and yielded a higher pineapple fiber content in the spun yarns. However, it was also found that using the same process on abaca fiber resulted in lower performance in terms of ends-down and percent yarn realization.

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Next steps could be testing the pilot-scale adoptability of acid prehydrolysis in pineapple fiber and determining the performance of the resultant yarns in terms of fabric production and finishing application.

Natural dyes and green colorants

TESTING THE VIABILITY OF LOCAL PLANT-BASED DYES
Exhaust natural dyeing by indigenous fabrics from selected Philippine plant dye sources (Year II)

Synthetic dyes have long replaced the traditional use of natural dyes in coloring fabrics. However, due to increased public awareness on environmental issues regarding synthetic dye production and application, interest is once again directed to the use of natural dyes. Natural dyes are extracted from natural resources including plants, insects, and minerals, and are non-toxic, eco-friendly, and sustainable alternatives to synthetic dyes and their harmful properties.

Using more than 100 natural dye resources, PTRI has continuously developed technologies on natural dye crude extraction, textile application, and dye powder production. This time it conducted a study which aimed to maximize the degree of exhaustion or uptake (the amount of dye which transfers and bonds with the fabric) of natural dyes by different fabrics. By conducting studies on the natural alternative, the Institute hoped to take large

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strides towards reducing water and energy consumption, creating a wider range of colors and shades, and maximizing natural dye resources.

Dyes were extracted from nine different Philippine dye-yielding plants: sappan bark, annatto seed, cogon grass, mayana leaves, yellow ginger, coffee pulp, coconut husk, talisay leaves, and mahogany bark. The plants were selected based on their availability and sustainability as identified in a previous project titled “Upscaling and Expanding Production and Application of the Philippine Natural Dyes.” The dyes were then tested on cotton, pineapple leaf fiber, and abaca fibers, on the basis of natural dye source, dye bath exhaustion, and exhaustion curve.

The study found that colorants can still be extracted from the source plants up to five or six times, with coconut lasting up to six extractions and the other materials lasting up to five. The strongest color strength values (K/S) were derived from the first to third extraction and, when applied, the dyes exhibit different properties depending on the fabric that it is applied to. The research was also able to determine that dye absorption was generally rapid in the initial stages of contact time and gradually decreased with time with equilibrium, and that recycling exhausted dye baths reduces water and energy waste while maximizing dye usage.

The dyed materials were also subjected to laundering and perspiration tests to determine colorfastness (resistance to fading and running). Cotton, abaca, and pineapple-silk fabrics which used dye from coffee pulp, mahogany, coconut husk, talisay, black, cogon, and mayana performed well. Meanwhile, fabrics which used dye from annatto, yellow ginger, and sappan exhibited a tendency to stain, with the last also resulting in a color change.

PTRI’S FIRST NATURAL DYE SWATCH BOOK
Color Profiling of Natural Dyes from the Established NatDyes Production and Dyeing Facilities for the Development of Natural Dyes Color System (Phase I)

Public concern related to health and environmental issues tied to the disposal of dyes, combined with government-issued mandates to address these issues, has increased the demand for sustainable colorants. Given the amount of dye wastewater being released into the ocean, governments, markets and industries are now scrambling to look for eco-friendly natural alternatives to synthetic dyes.

With the Philippines being a bio-diverse country with an abundant resource of botanical dyes, the PTRI officially established the NatDyes Center, a facility dedicated to everything involving natural dyestuff, natural dyed abaca-based products, and the selling of natural dyed yarns and fabrics.

To date, there are currently 74 documented botanical sources of natural dyes. While there is a large selection of natural resources, however, final dye colors are heavily dependent both on the source material, and the process that it is put through. There was thus a need to develop a system which profiles and catalogues the natural dyes.

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Adopting the Pantone color system, an internationally recognized method of classifying colors, the Institute set out to create a complete color profile of natural dyes made from resources that are present in the NatDye production hubs located in Abra, Mindoro, and Lake Sebu. The extracted dyes were then tested on pure knit cotton, pure abaca, handwoven pure pineapple, and handwoven pineapple silk, all of which were sourced from a local knitting company and a local handloom weaving firm.

From Namarabar, Abra, researchers collected mahogany barks, kapok barks, and tawa-tawa leaves. In Lake Sebu, they were able to create yellow hues from gathered talisay leaves and yellow ginger. Meanwhile, in Nagbacalan, Ilocos, the weaving cooperative that manages the natural dyeing facility had already ventured into the use of natural dyes at the time of the study. The cooperative used mahogany, mango, sampalok, talisay, annatto, cogon blades, and camahile, and so these were the materials that were gathered and brought back to the lab for testing.

By the end of the project, the natural materials from all three locations and their resulting dyes were successfully profiled and compiled into a swatch book. The book contains swatched fabrics with the description of their dye source, sampling location, dye concentration, and utilized mordant (substances used to affix color onto fabric). In addition to this, a standard and systematic labelling system for the dyed fabrics was also put into place. The book is currently available for public use at the Research and Development Division at PTRI. The project will begin its second phase in 2019 to include more samples and, after the second stage, the book will be made available on the PTRI website.

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Biofunctional materials and smart textiles

NO MORE "CLINGY" CLOTH
Anti-static finishing technology for silk
and poly-blend indigenous textile

A fabric's performance can go beyond aesthetic qualities. The integration of new technologies and multifunctional values into garments has paved the way for the creation of smart textiles, or materials that have the capability to respond to different environmental conditions or stimuli (mechanical, thermal, chemical, electrical, etc.). Garments made from these materials are geared to adapt or favorably react to certain conditions to the benefit of the end-user.

The PTRI has been developing such technologies through its Smart Textiles R&D program. Using organic and inorganic nanomaterials, the Institute has developed multi-functional fabrics that can be used in clothing, furniture, and other technical textile applications. These include materials that exhibit antimicrobial, UV-protecting, self-cleaning, and water-repellent properties.

Despite rapid developments and innovations, however, the program has encountered problems in working with synthetic and hydrophobic textiles due to their inability to allow electricity to flow through the material. This quality generates static which interferes with the

speedy processes of spinning and weaving and has the tendency to attract foreign particles unto the material. The fabric may also cling to wearers and induce electric shocks.

In response to these issues, the PTRI has developed an anti-static finishing technology for silk and poly-blend indigenous textiles. This can easily be applied using the conventional pad-dry-cure method, a process by which the textile passes through a water-based solution containing the finishing treatment. The anti-static agent, called lauric diethanolamide (LDEA), contains hydrophobic (water resistant) and hydrophilic (water absorbent) properties, and has the ability to alter the material's intrinsic electrostatic behavior. The resulting finishing is thermally stable (meaning it is not likely to grow uncomfortably hot or burst into flame), does not affect the color of the material, reduces the surface charge (lessening tendency to cling), can be easily handled and distributed, does not give off an oily or sticky feeling, does not interfere with other additives, and is cost effective.

MOSQUITO-REPELLENT CLOTHING
A Product of TELAPilipinas and the Smart Textiles Program of DOST-PTRI

With mosquito-borne illnesses causing about one million deaths every year, protection against mosquito bites has become a major health concern. Global warming has heightened the incidence of dengue, malaria, and zika virus which in turn has increased demand and investment in the mosquito repellent market. The Philippines in particular is known to have a high incidence of dengue, especially among children ages 5 to 14 years old (notably. roughly 33% of the Pilippine population is under 14 years old).

Under the Smart Textiles program, the PTRI was able to develop a mosquito-repellent fabric made from natural materials. The cotton-, abaca-, and pineapple-based cloth is applied with anti-mosquito finishing formulated from natural essential oils. The active ingredient which wards off the mosquitoes is released from the fabric over time; effectiveness of the repellent can last up to six days, after which the mosquito repellent can be re-applied.

Thus far, the mosquito-repellent cloth has proven effective against Aedis aegypti mosquitoes, which are the principal carriers of dengue. Field testing is currently being done to establish effective dosage, protection range (area), duration and rechargeability. In addition, the repellent will be tested against three other mosquito species, including morning and night biters.

The PTRI exhibited the smart textile in the form of baby clothes during the National Science and Technology Week. The mosquito-repellent apparel is intended to be initially marketed to middle- to high income-earning mothers residing in urban areas. The Institute is currently also looking into using the textile for other items including patches, lanyards, and pillowcases. There is great potential for mosquito-repellent textile products to be further developed and marketed here and abroad.

Commented [JP15]: This is a placeholder, kindly provide a project tied to the product.

Commented [JP16]: Please clarify, is this a mosquito-repellent clothing or a treatment solution? Can we say it is the clothing from natural fabrics in tandem with the repellent—which may signify that there is no other type of clothing to which this will be applicable? Please let us know.

Solution questions: How does one re-apply or re-treat the clothing? Will laundering lessen its effectiveness? Where can potential customers/buyers get this solution? How long is the shelf life of the solution?

Do you have a commercial partner already and when will/was this rolled out?

Commented [JP17]: The most vulnerable to dengue and other mosquito-borne diseases are in the low-income sectors. Is there a cost-efficient variant designed for the poor that you're also looking into?

Intellectual property

As a development strategy and to protect the work of inventors, the PTRI registers the inventions and process methodologies resulting from the R&D with the Intellectual Property Office. For the reporting period, three applications were filed.

Commented [JP18]: What is their status? Do we say that they have been approved?

TITLE	ABSTRACT	INVENTORS	DATE
PRETREATED SUGARCANE BAGASSE FIBERS FOR THE TEXTILE YARN PRODUCTION	The present utility model relates to pretreated sugarcane bagasse fibers for textile yarn processing obtained by the method comprising the steps of alkali degumming, chemical and mechanical softening of the sugarcane bagasse fibers, and opening and blending of the softened fiber prior to yarn spinning.	Engr. Ronald Pechera, Engr. Robert Casero, Engr. Eduardo Marin, Ms. Jeannie Lynn Perez, Ms. Nora Mangalindan, Ms. Nelia Baldon, Ms..Lucena Yap, Mr. Eduardo Valentino, Ms. Ernesto Dangaran, Mr. Agustin Bordallo	11/5/2018
COTTON-SUGAR CANE BAGASSE BLENDED RING SPUN YARN	The present utility model describes cotton-sugar cane bagasse blended ring spun yarn obtainable by the method comprising the steps of cutting crashed sugar cane bagasse fibers in a chamber for chemical processing, and then drying the fibers; blending the dried pretreated sugarcane bagasse fibers with cotton fibers; and converting the blended cotton-sugar cane bagasse fibers into ring spun yarns through	Engr. Henry Listano, Engr. Daniel Lavin, Engr. May Rico, Ms. Beatriz Roxas	11/5/2018
METHOD OF PRODUCING COTTON-SUGAR CANE BAGASSE BLENDED RING SPUN YARN	The present utility model describes cotton-sugar cane bagasse blended ring spun yarn and a method of producing the same. The method comprises the steps of cutting crashed sugar cane bagasse fibers into a desired staple length using a cutting machine; placing the cut crashed sugar cane bagasse fibers in a chamber for chemical processing and then drying the fibers; blending the dried pretreated sugar cane bagasse fibers with cotton fibers; and	Engr. Henry Listano, Engr. Daniel Lavin, Engr. May Rico, Ms. Beatriz Roxas	11/5/2018

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Commented [JP20]: Lists the same people above. Please verify that this is not a clerical error.

	converting the blended cotton-sugar cane bagasse fibers into ring.		

[Division 2] **Technical Services**

With its industry-leading facilities and expertise, the PTRI is equipped to provide technical services to clients from various sectors including schools, private companies, public agencies and local government units. Through the use of PTRI's physical testing and chemical testing laboratories, clients have the means to test raw materials, correct production problems, and further develop their products and processes. Thus, PTRI contributes to the improvement of quality and productivity in the various sectors of its clientele.

Laboratory Testing

Physical and chemical testing

The physical testing laboratory is dedicated to the mechanical testing of materials, including fabrics, carpets, and woven and nonwoven geotextiles (permeable technical fabrics used in civil engineering construction projects) from various clients. This 2018, the physical testing laboratory more than doubled their output (118.6% over the initial target).

Meanwhile, the chemical testing laboratory is dedicated to the chemical and microscopic testing of different fibers, yarns, fabrics and garments.

[Infographic 5] **Laboratory testing for 2018**

	Physical lab	*Chemical lab*
No. of quality tests	3,613	3,491
No. of samples	1,126	1,443
No. of clients	795	783
Income generated (PHP)	2,281,048.00* ** 118.6% over the initial target*	1,413,606.00

[End of Infographic 5]

Physical quality testing services amounting to 3,014 tests (83.4% of the total) were provided to 741 customers composed of textile millers, garment manufacturers, traders, government agencies, private offices, students and allied industries, generating an income of PHP 1.82 million for the government.

Meanwhile, a total of 56 customers composed of construction companies, manufacturers, suppliers, non-government organizations, government institutions, and academic institutions availed of geotextiles/geosynthetic testing services of the physical testing laboratory.

[Infographic 6] **New clients for 2018**

The majority of new clients for PTRI's laboratories belong to the MSME sector.

	Physical lab	*Chemical lab*
No. of new clients	**81**	**83**
MSMEs	56 (69%)	
Large companies	-	
Academe	22 (27%)	
Government	3 (4%)	

[End of infographic 6]

Among the more interesting tested materials is baconet, which is a geotextile made of banana fiber and coconut coir developed to prevent soil erosion and landslide.

Other notable projects include the development of uniforms for members of the Philippine National Army and the Philippine Navy, firefighter uniforms, as well as office uniforms for private and government offices, linens and laboratory gowns for the Philippine General Hospital, and inmate uniforms for the Bureau of Corrections and the Bureau of Jail Management and Penology.

The laboratories also tested packaging for Duty Free Philippines, and carpets and underlays for clients in the automotive industry.

Laboratory management

The Physical Testing Laboratory, which maintains 34 mechanical testing methods, was successfully accredited under ISO/IEC 17025:2005 (valid until November 20, 2022).

Nine newly acquired pieces of equipment have replaced the outdated ones in the office. These include a pilling resistance tester, three flammability testers, a high-pressure bursting strength tester, a thickness tester for geotextile, a thickness tester for fabric, and a wrap reel. Laboratory personnel and support staff underwent training on the operation of these new equipment and received satisfactory results from the ATSM Proficiency Testing Program, which assessed their ability to competently perform specific tests.

In compliance with the Chemistry Profession Act (Republic Act 10657), the Chemical Testing Laboratory underwent evaluation under the Professional Regulation Commission and the Professional Regulatory Board of Chemistry. After the facility was deemed up to standard, the PTRI received the certificate of authority to operate (Certificate No. 80, valid until August 30, 2021). In addition, the facility has continually improved its laboratory management system—submitting its GAP Analysis or laboratory assessment readiness program to the Philippine Accreditation Bureau in July 2018, and successfully conducting an internal audit in September 2018—in accordance with the upcoming transition from PNS ISO/IEC 17025:2005 to PNS 17025:2015 this 2019.

Commented [JP21]: May change depending on chemical lab information. Also: It depends on whether you really wish to highlight this data. Please let us know your preference.

Commented [JP22]: Information provided did not present the data in similar fashion as physical testing lab. Please be so kind as to re-classify data for comparability.

The intention is to show the number of MSMEs who benefited from PTRI's services insofar as new clientele is concerned.

If you wish a different presentation and focus, please be so kind as too provide the necessary data.

Commented [JP23]: For BFP or for PITC?

Similarly, the Chemical Testing Laboratory upgraded its facilities, including the chemical stockroom, and the chemical filtration area, and the soxhlet extraction area. Eight pieces of laboratory equipment underwent calibration and preventive maintenance. Equipment upgrades under ONELAB also included a forced-draft oven, Farnsworth Mansell 100 Hue Test Kit, electronic crockmeter, top loading balance, dessicator cabinets, and dessicators without stopcock.

Textile processing and technical services

New weave designs

PTRI developed 78 brand-new weave designs for home decors, bags and handicrafts. While the designs were originally created for the purpose of a student thesis, the patterns were later adopted by several weaving associations based in Guinzadan, Mountain Province, the Butterfly Haven in Pulilan, Bulacan, and the Negros Nine Human Development Foundation Inc. for their own creations.

Weaving innovations

Other accomplishments for the year included the development of various apparatus for weaving. One is the modified therapeutic loom, which only requires eye-hand coordination. Dubbed as the *thera loom*, it was created with to suit persons with disability, whether they wish to take up weaving for recreation or livelihood. Another accomplishment is the modified sectional warping machine with a detachable warp beam, which is designed for a continuous flow of the handloom weaving process. Lastly, with the goal of improving productivity, the Institute was also able to develop an improved version of the handloom which has a semi-automatic foot switch system.

The PTRI was also able to accredit the loom fabricator of Danilyn's Enterprises, Inc. after ensuring that their processes and equipment were up to PTRI standards.

Other technical services

The PTRI Innovation Center for Yarns and Textiles aided various clients in their small and laboratory scale processing requirements. These services involved spinning, twisting, powerloom and handloom weaving processes, and colormatching and dyeing services.

A total of 926 technical services were provided to 66 clients involved in the manufacture and research of textile and allied products using raw fibers (blended or otherwise).

[Infographic 7] **Technical services per client, 2018**

MSMEs	443 services (48%)
Academic institutions	290 services (31%)
LGUs	193 services (21%)

[End infographic 7]

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Consultancy Services

A total of 473 customers consulted with the Physical Testing Laboratory, whereas 322 customers consulted with the Chemical Testing Laboratory. PTRI also provided 902 consultancy services to [x number of] customers on textile processing and technical specifications.

Consultancy services are provided to a cross-section of private sector, public sector, and academe, including complimentary services given to students who approached the Institute for help with their thesis or science projects.

PTRI's consultancy services aim to assist the various sectors involved in textile research and textile production, particularly in evaluation, conformance to specifications, and troubleshooting techniques.

In general, topics for consultation involve textile processing, utilization of indigenous fibers, handloom specifications and fabrication, weave designs, twining/twisting machines, fiber extraction machines, mildew prevention, water repellency, color formulation, dye stripping, re-dyeing, dye fixing, shrinkage control, and other production-related concerns.

Specific concerns this year included the proper selection of beddings/linens and staff uniform for the Philippine General Hospital and the Philippine Heart Center, among others, and the properties of firefighter uniforms as queried by the Philippine International Trading Center and the Bureau of Fire Protection.

[Infographic 8] **Customers assisted in 2018**

MSMES	
The PTRI was able to visit 22 MSMEs to assist them in their production. They also looked into possible collaborations between the institution and the different stakeholders	
Weaving technology utilizing natural textile fibers for tent	Joeus Event Supplies Corp. (Sampaloc, Manila)
?	Medtecs International Corp. Ltd. Factory Plant (Mariveles, Bataan)

Schools and Universities
- St. Paul University, Manila - University of Sto. Tomas - Lyceum University of the Philippines - Don Mariano Marcos Memorial State University - Technological University of the Philippines

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Also: If people wanted to partner with you, what information will we need to provide? We can add a footnote or blurb for that.

Commented [JP26]: Both are mentioned. BFP is understandable, but why PITC?

Commented [JP27]: The complete list was not provided. Please be so kind as to complete the list—if this data is important to you.

Another way of presenting data, should you wish to provide us with the numbers:

of MSMEs assisted
of students (broken down into level) assisted
of government units assisted
etc.

A list of the types of assistance provided per client category will also be informative.

Understandably, you have a lot of customers and many interventions – we can be selective in terms of which are the most impactful?

- Eulogio “Amang” Rodriguez Institute of Science and Technology - Makati Science High School - Far Eastern University - University of the East Ramon Magsaysay Memorial Medical Center - Nursing	
Dyeing technology utilizing various types of softening agent (sansoftener CA1; Tubingal 6069; AS10) for handwoven fabrics made from cotton/snake plant	Makati Science High School
Textile processing technology using decorated fibers from maguey and snake plant	Abra National High School

Government Units	
Dyeing of Indigenous Fibers	Local Government Unit, Calamba

[End of Infographic 8]

[Division 3] **Technology Transfer, Information, and Promotion Services**

As the main proponent of textile development, the PTRI is focused on serving the industry and the nation by imparting its knowledge and expertise. Believing in the potential of textiles in fueling development and empowering individuals, it aims to strengthen the competitiveness and competencies of MSMEs and other beneficiaries, to provide information about the industry and its developments to the public, and to inspire every Filipino to take an interest in local textiles. These are done through technology transfer, information services, and promotion services.

Technology Transfer

LOCAL INDUSTRY GIVEN BOOST THROUGH SERICULTURE

Support to the Development of the Mindanao Silk Through the PTRI Technology Center, Misamis Oriental (TCMO)

As part of the initiative to reviving the sericulture industry in Mindanao, the PTRI-Technology Center in Misamis Oriental (TCMO) set out to strengthen manpower of the center and enhance overall technical competencies.

The first phase of the project included the integration of seven new and younger personnel to the center. The introduction of younger staff members was crucial given that the staff lead on silkworm rearing was nearing retirement age. The second stage involved building on the competencies of the staff. In its 40 years of the center's service, none of the four staff members had received any formal training on silkworm plasm maintenance and development. To ensure that knowledge and competencies exhibited by staff were adept and up-to-date, technical staff members were sent to train with internationally known centers and institutions to learn and unlearn the practices in silkworm rearing and familiarize themselves with new organic methods and dyeing techniques. Their training included lectures, workshops, and area visits to production sites.

Program	Date/Venue
Methods of Maintenance of Silkworm Germplasm Resources	September 18 - October 5 / Yamanashi, Japan
Training in PDR Lao	November 17-23 / PDR Lao
Sericulture and Silk Industry	December 2-29 / India
World Ikat Textiles Symposiums 2018	December 4-6 / Khon Kaen, Thailand

Echo symposiums were then conducted to relay the information gathered from the trainings. The symposiums were attended by the technical team in TCMO and the project partners.

The project was also able to upgrade the germplasm and rearing facilities, increase mulberry propagation of the center, and strengthen the facility's overall capabilities. In addition to the existing 2 hectares of rented land for sericulture, 2.5 hectares were newly dedicated for planting mulberry trees. A rearing house that can accommodate two boxes of silkworms was also built. Meanwhile, the offices and the laboratory were also upgraded and granted the necessary supplies. Mobility capabilities were also improved with the acquisition of a 4x4 vehicle and a motorcycle.

Lastly, the PTRI successfully established a micro-scale silk reeling and throwing facility by revamping an old loom-weaving building. Improvements were made to meet structural standards. The TCMO is also set to install the necessary equipment including a boiler, a reeling and re-reeling machine, a cocoon dryer and a silk throwing facility.

Commented [28]: Were they able to install this?

In addition to this, the PTRI also provided technical services and initial mulberry cuttings to three potential partners: the women's group in Barangay Balubal under the City Housing Urban Development Department of Cagayan de Oro City, the farmer's group under the City Agricultural Office of Malaybalay City, and the farmer's group under Barangay Malagana in Claveria, Misamis Oriental.

With all of these new developments, the TCMO is expected to increase its cocoon production by 94%, and produce 1,344kg of raw silk annually.

Support for Productivity Enhancement of Sericulture, Handloom Weaving and Natural Dyeing in Negros Occidental

Through the efforts of the Oisca Bago Training Center (OBTC), Negros has long been famous for contributing no less than 75% of the silk in the country. However, recent years have shown a decline in the quality and quantity of their silk, which is attributed to the poor performance of the degraded reeling facility and/or the poor quality of the hybrids. In light of this, the PTRI was determined to revive and improve sericulture productivity in the Visayas.

In building the capabilities of the OISCA-BTC, it was important to invest in staff training. One technical staff, along with three other members of the PTRI, was sent to Thailand to attend the World Ikat Textiles Symposium 2018 which included seminars, workshops and area visits to further hone their knowledge on silkworm rearing and maintenance. Knowledge acquired through the seminars were then passed onto the other technical staff members through an echo seminar.

The project also evaluated the performance of silkworms within the facility and the properties of their resulting silk. Observing the silkworms during in their egg stage, larval stage, and cocoon stage, the following were studied: hatching percentage, larval rate, larval duration, larval mortality, numbers of cocoon per liter, weight of cocoons per liter, single cocoon weight, cocoon shell weight, shell percentage, live pupa percentage, and the total reel-able cocoon weight.

Lastly, the project supported the upgrade of the cold storage facility and the establishment of a brand new throwing facility. The household refrigeration unit being used

to preserve silkworm eggs was replaced with an upgraded cold storage unit in order to provide the optimal conditions for preserving silkworm eggs. The Institute also plans to provide the center with a new throwing facility.

Commented [29]: Did this push through?

[Standalone sidebar] ***"Thera transfer"***
DOST provides therapeutic handlooms to differently-abled and elderly

With the PTRI's commitment to uplifting lives through textile innovations, the institute was able to provide 10 units of modified therapeutic handlooms, dubbed *thera looms,* for the differently abled and the elderly of the Guanella Center, Inc. in Pasong Tama, Quezon City.

Unlike traditional handlooms that require rigorous effort in order to create the cloth, the PTRI-made handlooms were modified so that the user would only need hand-eye coordination, making it possible for the elderly and the differently-abled to use them for leisure and as a means of livelihood. The looms were designed for durability and smooth operation while increasing overall efficiency and productivity even for those who have no previous experience in weaving.

Guanella Center, Inc. is a nonprofit run by the Servants of Charity, whose mission is to provide the poor and disabled charitable services and assistance, including sheltering and educating persons with disabilities. They are an ideal beneficiary for the thera looms, which can be used in their skills training and livelihood programs for the PWDs and elderly.

[End of sidebar]

Skills and Knowledge Transfer

PTRI is dedicated to strengthening the textile industry by honing the knowledge and competencies of the different stakeholders of the industry, and empowering communities who could potentially part of the network. One of the ways in which this mission is attained is by conducting various training courses on industry-related topics.

Province	Intervention	Name of Beneficiary
CAR		
Mountain Province	Basic Course on Dyeing of Indigenous Fibers	Guinzadan National High School
		Tadian Municipal Weavers Organization
Kalinga	Advanced Handloom Weaving Technology	KINWA
		DARAM Multipurpose Cooperative
Mountain Province	Extraction and Application of Natural Dyes	Guinzadan National High School
		Bauko Weavers & Sewers Association (BAWSA)
		Guinzadan Mam-abe Organization

		Tadian Municipal Weavers Organization
		Inabe Livelihood Group Inc.
		Guinzadan National High School
		Guinzadan Weavers
		Inabe Livelihood Group Inc.
Mountain Province	Basic Course on Dyeing of Indigenous Fibers	Tadian Municipal Weavers Organization
		Sabangan Weaving Association
		Bauko Weavers & Sewers Association
		Inabe Livelihood Group Inc.
		Namatec Weaving Organization
		Guinzadan Man-abe Org.
Mountain Province	Ikat Weaving Technique	TAMWO
REGION I		
		Arjaycee Craft
		Association of Maize Crafters de San Juan
		Bauang Crochet Association
		DMMMSU-SRDI
		D'Senos Creations
		EDC Paperhaus & Gen. Sales
		ELYU Trading
		Northway Arts & Crafts
		Owner Guapple & Grapes Grower/ Parties
		Paringao RIC Food Terminal
		Santol Handicraft Business Center
		Sericulture Handicrafts Producers Association
La Union	Extraction and Application of Natural Dyes	Silk Weaving Association

REGION II		
Nueva Vizcaya	Advanced Handloom Weaving Technology	Alayan Pag-asa Abot Palad Association
REGION III		
Bulacan	Basic Course on Dyeing of Indigenous Fibers	PUNIQUE Handicrafts
Bulacan	Advanced Handloom Weaving Technology	Handloom Weavers of Pulilan
REGION IV-A		
Batangas	Extraction and Textile Application	Natural Craft Connection Enterprises
Batangas	Natural Dye Powder Production	Natural Craft Connection Enterprises
Laguna	Basic Course on Handloom Weaving Technology	Likhang Kamay ng Kalambenyo Producer's Association
Laguna	Seminar/Workshop on Synthetic Dyeing of Indigenous Fibers	Likhang Kamay ng Kalambenyo Producer's Association
Laguna	Advanced Handloom Weaving Technology	TAYO Foundation
Cavite	Extraction and Application of Natural Dyes	Sanctuario Nature Farms
Taguig	Basic Course on Handloom Weaving Technology	Technological University of the Philippines
NCR		
Taguig	Physical Quality Textile Testing	Irvine True & Frank Carson Philippines
Parañaque	Extraction and Application of Natural Dyes	Creative Definitions General Merchandise
Manila	Quality Textile Testing	TUP-Manila
Parañaque	Advanced Handloom Weaving Technology	Creative Definitions General Merchandise
Quezon City	Basic Course on Handloom Weaving Technology	Guanella Center Inc. Servants of Charity Elderly Ministry
REGION V		
Sorsogon	Basic Course on Dyeing of Indigenous Fibers	Chuwan's Handicrafts Native Slipper Makers Salanigo Weavers
REGION VI		
Negros Occidental	Extraction and Application of Natural Dyes	Negros Nine Human Development Foundation Inc.

Negros Occidental	Advanced Handloom Weaving Technology	Negros Nine Human Development Foundation Inc.
Aklan	Basic Course on Handloom Weaving Technology	Guadalupe Handicraft Association
REGION VIII		
Biliran	Basic Course on Dyeing of Indigenous Fibers	Cabucgayan Agrarian Reform Cooperative (CARECO)
REGION X		
Misamis Oriental	Basic Course on Dyeing of Cellulosic Fibers	Abai Weavers Multipurpose Cooperative
Misamis Oriental	Degumming Technology for Thrown Silk/Yarns	Abai Weavers Multipurpose Cooperative
REGION XII		
South Cotabato	Basic Course on Handloom Weaving Technology	Kenhulung Handicrafts Federation Inc.
Surigao del Norte	Extraction and Application of Natural Dyes	Socorro Weaving

[Sidebar] **Innovation hubs**

The PTRI also transfers knowledge by way of innovation hubs and development centers designed to serve the local industry and academe.

PTRI NatDyes Center

The NatDyes Center in Bauan, Batangas was created to serve as the main facility for commercializing natural dye production for textile application. In 2018, the center served as the main base for a project that looked into cataloguing the natural dyes of the Philippines.

"Perfect fit TELA concept"
(Textile product development centers)
NEED INFO

"TELA yarn production and innovation center in Miag-ao, Iloilo"
NEED INFO

[End of sidebar]

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We will need the history of each, the services they provide, and their impact on the local community. What makes them relevant in terms of technology transfer, information dissemination, and promotion of indigenous textiles?

Commented [31]: NEW ADDITION

Information Services

Forum and symposium on sustainable production

As a response to the environmental and social issues coming to light in the textiles and manufacturing industries, PTRI embarked on two initiatives to promote sustainability in the workplace. On July 19-20, 2018, the PTRI launched the "Forum on Waste = Resource to Inclusive Circular Economy," which aimed to educate various stakeholders about a circular economy and the sustainable use of resources. The multi-sectoral event also gave the attendees from different sectors the opportunity to create new networks and partnerships that will enable them to collaborate on responsible sourcing, waste reduction, fair trade and labor, among other possible solutions that will directly address issues within their sector.

Following the success of the Forum on Waste, the PTRI also had the privilege to host the 5th InnoSight International Conference, an event co-organized with Taiwan's Industrial Technology Research Institute (ITRI). Dedicated to championing sustainability for the textile industry in the Asia-Pacific, speakers and experts from different parts of the world shared their experiences, insights, and expertise on various sustainability topics. The event also allowed participants to discuss the challenges and opportunities that they continue to face in the pursuit of sustainability, particularly how to curb fast fashion and mitigate pollution.

Promotional Events

Design competition for government uniforms

In an effort to integrate local weaves and natural textiles into modern everyday use and encourage public awareness and participation on developing local textiles, the PTRI proudly launched the 2018 Government Uniforms Design Competition.

Participants, composed of both experts and amateur designers competing in separate categories, were tasked to use local weaves and natural textiles to design uniforms for the Department of Science and Technology (DOST), Department of Foreign Affairs (DFA), Department of Tourism (DOT), Civil Service Commission (CSC), Department of Budget Management (DBM), and the Department of Education (DepEd).

RESULTS OF THE COMPETITION.

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[Division 4] Administrative Services

The PTRI continually strives for excellence in both its external and internal activities. This presupposes that all of its administrative staff will be motivated and capacitated to serve. To be active contributors in achieving the Institute's mission and vison, they must be up-to-date on the knowledge and skills needed in the industry they serve.

Training and motivation

General Assembly

- The Institute kicked off the year with their first general assembly on [place] at [date].

Team Building

- To strengthen the moral and camaraderie of the different members of the Institute

Tree-planting

- As part of their obligation to the preservation of the environment, [n] staff members for the PTRI partook in a tree planting project at [location]. A total of [x] trees were planted in a span of [n] days.

Competency Building

The members of the staff and administration attended various local and global training programs which focused on building their technical knowledge, skills, and capabilities. These training opportunities may sometimes be unplanned, being organized by external entities, but considered significant to manpower development and thus plotted into the calendar as the invitations or call-outs occur.

The training conferences, workshops, seminars, and lectures are as follows:

Trainings and Seminars (By Request)

January 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
17	4 PTRI Employees	Budget Fora on the FY 2019 Budget Preparation	DBM/PICC Pasay City	
16-19	4 PTRI Employees	Hands-on Training on Solidworks 2017	CADCENTRIC Solutions Inc./Madrigal Business Park, Alabang, Muntinlupa City	
18	2 PTRI Employees	Technology Transfer and Selection Process of ICorp candidate/ready technologies	DOST/ADMATEL Bldg., DOST Compound, Bicutan, Taguig City	
23	11 PTRI	Orientation Training on	PTRI (PICTS)/PTRI	

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	Employees and 1 Contract of Service Member	Tracking, Retrieval, and Archiving of Communications for Efficiency (TRACE)	Training Room, Bicutan, Taguig City	
26	1 PTRI Employee	Tax Reform for Acceleration and Inclusion (TRAIN)	TRAIN/Millenia Suites, Ortigas, Pasig City	P3,359.44

February 2018

Duration	**Participants**	**Course/Title**	**Sponsor/Venue**	**Fees**
20 & 27	2 PTRI Employees	LEAP IP 2018 (Learn, Be Empowered, Adopt and Profit from IP) Seminar on Patents, Utility Models and Industrial Design	Intellectual Property of the Philippines (IPOPHL)/Intellectual Property Center, #28 Upper McKinley Road, McKinley Town Center, Fort Bonifacio, Taguig City	
19	2 PTRI Employees	Symposium on the Updates of the Chemistry Profession Act, A Chemistry Week 2018 Symposium	Integrated Chemists of the Philippines (ICP)/Century Park Hotel, Manila	P5,000.00
19	1 PTRI Employee	Innovation Forum	DOST/Philippine Institute of Volcanology and Seismology (PHIVOLCS) Auditorium, C.P. Garcia Ave., Diliman, Q.C.	
19-21 22-28	2 PTRI Employees	Filipinnovation Entrepreneurship Corps Lean Startup Training Program - Kickoff Workshop - Week 1 Customer Discovery	DLSU-RTI International-George Washington University/EDC Room, Henry Sy Bldg., DLSU, Taft Ave., Manila	
21-24	4 PTRI Employees	79TH PIChe National Convention with the theme *"Expanding Chemical Engineering Competencies for National Development"*	DOST-Philippine Institute of Chemical Engineers (PIChe)/Bellevue Manila, North Bridgeway, Filinvest City, Alabang, Muntinlupa City	P6,500.00
21-22	1 PTRI Employee	OneLab Project Coordinators' Workshop	PCIEERD/Cebu City	

March 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
14-16	Richard G. Apuyan	In-house Training on the Effective, Efficient and Compliant Records Management in the DOST System	DOST/Platinum Auditorium Bldg., MIRDC Compound, Taguig City	
1 8 15 16 2-7 9-14	Angelito T. Uldo Zailla F. Payag	Filipinnovation Entrepreneurship Corps Lean Startup Training Program - Lessons Learned Webinar #1 - Lessons Learned Webinar #2 - Lessons Learned Webinar #3 - Next Steps Webinar #4 - Week 2 Customer Discovery - Week 3 Customer Discovery	DLSU-RTI International-George Washington University/EDC Room, Henry Sy Bldg., DLSU, Taft Ave., Manila	
3	1 PTRI Employee	"Tatak Pinoy: Filipino Innovations"	The Mind Museum/JY Campos Park, 3rd Ave., Bonifacio Global City, Taguig	
8	2 PTRI Employees	Forum on International Cooperation: Opportunities for Science, Technology and Innovation (STI) Initiatives" and "Briefing on Public Investment Program (PIP) Updating Guidelines and Hands-on Session on the PIPOL System"	DOST/Platinum Auditorium, Platinum Building, MIRDC Compound, Bicutan, Taguig City	
9	2 PTRI Employees	"Training Workshop on the Use of the 2017 Agency Procurement Compliance Performance Indicators (APCPI) System"	Government Procurement Policy Board-Technical Support Office (GPPB-TSO)/Mt. Everest Function Room, 8th Floor, BSA Twin Towers Hotel, Doña Julia Vargas Ave. cor. Bank Drive, Ortigas Center, Pasig City	
20	1 PTRI	LEAP (Learn, Be	Intellectual Property of	

	Employee	Empowered, Adopt And Profit) 2018 Free Introductory Seminar on Intellectual Property (IP) – Topic: Patent	the Philippines (IPOPHL)/IPOPHL Multipurpose Hall, Taguig City	
20-23	1 PTRI Employee	Southeast Asia Business Studies (SEABS) University of Asia and the Pacific (UA&P) Leadership Development Program (LDP) Module 1 – Growing as a Leader	SEABS-UA&P/Pacific Pearl Drive, Ortigas Center, Pasig City	P50,000.00
22	1 PTRI Employee	Science for Change Program	DOST-PCIEERD/2nd Floor, Function Room 5, SMX Convention Center	
23	23 PTRI Employees including Contract of Service Members	DOST Women's Month Celebration with the theme "We Make Change Work for Women"	DOST/Philippine Science High School (PSHS) Gymnasium, Agham Road, Diliman, Q.C.	
24	9 PTRI Employees	PTRI Women's Month (Conduct of Zumba)	PTRI/PTRI Lobby	
26-28	1 PTRI Employee	Media Core Summit	STII/B Hotel, Alabang City	

April 2018

Duration	**Participants**	**Course/Title**	**Sponsor/Venue**	**Fees**
4-6	2 PTRI Employees	3-day Seminar-Workshop on Disaster Preparedness and Mitigation	Philippine Science High School (PSHS)/Subic, Zambales	
13	1 PTRI Employee	Innovation for Inclusive Impact (In_Pact) Asia Forum 2018: "Designing Tomorrow's World Today"	In_Pact Asia Forum 2018/5F, Multipurpose Hall, Henry Sy Building, DLSU, 2401 Taft Ave., Manila	
24-27	1 PTRI Employee	2018 Career Executive Service (CES) Executive Leadership, Wellness and Work-Life Balance Camp	CESB/Green Canyon Eco Art Resort, Tarlac	P14,000.00
26	1PTRI Employee	Intellectual Property Convergence	Intellectual Property of the Philippines (IPOPHL)/Rigodon Ballroom, The Peninsula Hotel, Makati City	

May 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
8	1 PTRI Employee	ISO/IEC 17025:2017 Awareness Seminar	OneLab/PCIEERD Conference Room	
8	3 PTRI Employees	ISO/IEC 17025:2017 Awareness Seminar	OneLab/PCIEERD Conference Room	
11	1 PTRI Employee	Seminar on Weathering	Gryke Scientific Instruments Trading/Acacia Hotel Manila, Alabang, Muntinlupa City	
22-25	1 PTRI Employee	Southeast Asia Business Studies (SEABS) University of Asia and the Pacific (UA&P) Leadership Development Program (LDP) Module 2 – Fundamentals of Business	SEABS-UA&P/Pacific Pearl Drive, Ortigas Center, Pasig City	Part of P50,000.00
23	4 PTRI Employees	Seminar on "Health, Nutrition and Fitness for Older Adults"	Food and Nutrition Research Institute (FNRI)/FNRI Training Room, Bicutan, Taguig City	
23 onwards for a period of 27 weeks	1 PTRI Employee	Online Training on *"Cybersecurity Specialization of the University of Maryland"*	DOST-GIA Project *"Learning at Scale Vol. 1: Data Science Track Extension"*	
23-26	1 PTRI Employee	40th GACPA Annual Convention and Seminar with the theme *"Sustaining Unity in Nation – Building through Service, Harmony and Innovation to Nurture Economy (SUNSHINE)"*	Government Association of Certified Public Accountants (GACPA)/Waterfront Hotel, Lahug, Cebu City	P8,000.00
24-25	1 PTRI Employee	Philippine Accreditation Bureau (PAB) Assessors and Experts' Forum 2018	PAB-Department of Trade and Industry (DTI)/Jade Hotel and Suites, 1120 Don Chino Roces Avenue, Barangay La Paz, Makati City	
26	1 PTRI Employee	"MUNI Meetup Mnl on a Plastic Ocean"	MUNI/Acceler8, 111 Paseo de Roxas, Legaspi Village, Makati City	P500.00
30-1 June	1 PTRI Employees and 2 Contract of Service Members	33rd Philippine Chemistry Congress with the theme: *"Chemistry: Rising to Challenges"*	Integrated Chemists of the Philippines (ICP)/PICC, Pasay City	P45,000

30-1 June	2 PTRI Employees	33rd Philippine Chemistry Congress with the theme: *"Chemistry: Rising to Challenges"*	Integrated Chemists of the Philippines (ICP)/PICC, Pasay City	P8,500.00 P8,500.00

June 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
6-7	1 PTRI Employee	Workshop on e-Onelab Progress Review	OneLab/Bohol Bee Farm, Panglao, Bohol	
8	Some PTRI employees including	Awareness Seminar on ISO 9001:2015	MIRDC/PTRI Training Room	P15,000.00
	Contract of Service Member			
27-29	2 PTRI Employees	Training Course on Monitoring and Evaluation	DOST/Conference Room 8,DAP Conference Center, Tagaytay-Calamba Road, Barangay Sungay East, Tagaytay City, Cavite	
20	1 PTRI Employee	Leadership Series Training on Digital Leadership	Civil Service Institute (CSI)/Crown Plaza Manila Galleria, 1100 Ortigas Ave., Ortigas Center, Quezon City	P5,000.00
18	2 PTRI Employees	Internet of Things with Blockchain for Security and Trust: Technologies, Applications, Benefits, and Challenges	DOST-ASTI-UP EEEI/ASTI Training Room	
16	3 PTRI Employees	MUNI Meetup Mnl on Sustainable Fashion	MUNI/Clock In High Street, C2 Bldg., 7th Ave., Bonifacio Global City, Taguig City	P500.00 P500.00
26-27	6 PTRI Employees	Training on the Harmonized Gender and Development Guidelines (HGDG)	DOST/Platinum Bldg., MIRDC, Bicutan, Taguig City	

July 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
5-6	1 PTRI Employee and 1 Contract of Service Member	Women's Economic Empowerment (WEE) Project Inter-Agency Consultation Workshop	Philippine Commission on Women (PCW)	
24-26	2 Contract of Service Member	Training Course on Intellectual Property Law,	ITDI/ITDI Standards and Testing Division	

Commented [35]: Ma. Eugenia A. Ablan
Richard G. Apuyan
Alexander M. Argame, Jr.
Charles Oliver C. Avenido
Josanelle Angela V. Bilo
Agustin A. Bordallo, Jr.
Teresa T. Canonce
Jeremias V. Caras
Melody Y. Laxamana
Henry R. Listano
Cornelio V. Loyola, Jr.
Joanna Tess M. Manuel
Mae F. Roa
Josue C. Rosal
Ronelo P. Vallente
Norly B. Villar
Kimberly P. Viron

Commented [36]: Marisol C. Gammad
Ma. Corazon S. Alviar
Heherson Alembert A. Zosa
Angelyn S. Idala
Mylene M. Soriano
Noriel G. Sorsone
Ma. Teresa F. Javier
Carlo M. Macaspag
Enrique E. Villanueva
Andres Jose L. Remoroza
Samilo O. Garcia
Freyzinet Charm C. Gonzales
Dioni R. Poblete, Jr.
Sharmainne Rhey B. Caoili
Jeramie A. Estoque
Dianne C. Noguera
Ephraim H. De Gracia
John Eduard S. Solahis
Villa Lee E. Delima
Dannieline M. Solis
Emmanuel Victor D. Buniao

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		Patent Search and Writeshops	Conference Room, Bicutan, Taguig City	
24-27	1 PTRI Employee	Southeast Asia Business Studies (SEABS) University of Asia and the Pacific (UA&P) Leadership Development Program (LDP) Module 3 - Optimizing Strategy	SEABS-UA&P/Pacific Pearl Drive, Ortigas Center, Pasig City	Part of P50,000.00
26	2 PTRI Employees	Cash Planning Report Forum	DOST-Bureau of Treasury (BTR)/Marble Hall, Bureau of the Treasury, Ayuntamiento de Manila, Intramuros, Manila	

August 2018

Duration	**Participants**	**Course/Title**	**Sponsor/Venue**	**Fees**
30 July - 3 August	1 Contract of Service Member	Prototype Designing Applicable to DOST Researchers using Autocad 2D/3D with 3D Printing	ITDI/Metrology Conference Room, Metrology Bldg., DOST Complex, Gen. Santos Ave., Bicutan, Taguig City	
14-15	1 PTRI Employee	Third (3rd Quarter) Cluster Meeting for CY 2018 of agencies under the jurisdiction of CSCFO-DOST	CSC-NCR/Bureau of Corrections (BUCOR) Conference Room, BUCOR Bldg., New Bilibid Prisons Complex, Poblacion, Muntinlupa City	
16-18	3 PTRI Employees	Advanced Materials and Nanotechnology Lecture and Workshops (lecture and workshops)	Ateneo de Manila University, Quezon City	
23-24	1 PTRI Employee	e-OneLab Planning Workshop for Y3 Project Implementation	Bacolod City	

September 2018

Duration	**Participants**	**Course/Title**	**Sponsor/Venue**	**Fees**
3 - 7	2 PTRI Employees	Specialized Training Course on “Organizational Development: Enhanced Readiness for Tomorrow’s	MIRDC/Titanium Auditorium, Titanium Bldg.,MIRDC, Bicutan, Taguig City	

		Challenges"		
5	1 PTRI Employee	One-Day Symposium on R&D Management: Discipline and Practice	UP Open University (UPOU), Los Baños, Laguna	
5 - 6	1 PTRI Employee	Leaders in Innovation Fellowships Event	Asian Institute of Management (AIM), 123 Paseo de Roxas, Legaspi Village, Makati City	
11-14	1 PTRI Employee	Southeast Asia Business Studies (SEABS) University of Asia and the Pacific (UA&P) Leadership Development Program (LDP) Module 4 – Leading Sustainable Change	SEABS-UA&P/Pacific Pearl Drive, Ortigas Center, Pasig City	Part of P50,000.00
17 19-20	1 PTRI Employee	Effective Leadership Skills Problem Solving and Decision Making Techniques	Philippine Trade Training Center (PTTC)/Roxas Blvd., Pasay City	P2.500.00 P3,000.00
18	2 PTRI Employees	DOST-3M MOU Implementation Planning Workshop	DOST/3M Customer Technical Center, Cocolight Bldg., 11th Ave., cor. 39th St., Bonifacio Global City (BGC), Taguig City	
18 - 20	2 PTRI Employees	Workshop on Capability Building on Policy Translation and Advocacy: Empowering R&D and S&T DOST Staff	FNRI/FNRI Training Room, 3rd Floor, FNRI Bldg., Gen. Santos Ave., Bicutan, Taguig City	
20 - 21	1 PTRI Employee	67th PHILAAST Annual Convention with the theme: *"Engaging the Youth in Science, Technology and Innovation"*	Philippine Association for the Advancement of Science and Technology (PHILAAST)/PHIVOLCS Auditorium,C.P. Garcia Ave., Diliman, Quezon City	P2,500.00z
20-21	4 PTRI Employees and 1 Contract of Service Member	International Design Conference 2018	Design Center of the Philippines/Ballroom A, Ascott, Bonifacio Global City, 5th Ave., cor. 28th St.,BGC Taguig City & Samsung Hall, SM Aura Premier, Taguig City	
21	6 PTRI Employees and 1 Contract of Service Member	Awareness Seminar on Transition from ISO/IEC 17025:2005 to ISO/IEC 17025:2017	Accuservices, Inc./Berjaya Hotel Makati, 7835 corner Eduque St., Makati Ave., Makati City	P14,000.00 (OneLab)

21	1 PTRI Employee	ISO/IEC 17025:2017 Awareness Training	Accuservices/Berjaya Hotel, 7835 cor. Eduque St., Makati Ave., Makati City	P2,000.00 (OneLab)

October 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
8	3 PTRI Employees	First Stakeholders' Forum on Nanotechnology	ITDI/Muralla Room 2 & 3, Bayleaf Hotel, Intramuros, Manila	
9 – 12	1 PTRI Employee	Supervisory Development Course (SDC) Track 1	CSC-NCR/Quezon City	P12,000.00
11	1 Contract of Service Member	Social Media Engagement Workshop	PCHRD-Project Management Office-Science for Change Program (PMO-S4CP)/Lecture Room, 2nd Floor, ADMATEL Bldg., DOST Compound, Bicutan, Taguig City	
17	1 PTRI Employees and 2 Contract of Service Members	"Innovation in Thermal Conductivity, and Raman Microscopy for Advanced Material Testing"	ITS-C-Therm Technologies-Renishaw/Admatel Lecture Room, DOST Compound, Gen. Santos Ave., Bicutan, Taguig City	
22-23	2 PTRI Employees	"European Research Funding and Collaboration Opportunities: Horizon 2020 Info Days"	DOST-European Commission/University of the Philippines BGC, 14th Drive, University Parkways, Fort Bonifacio Global City, Taguig City	
22-24	1 Contract of Service Member	Three-Day Seminar-Workshop on Handling Media Interviews	DOST-FNRI/FNRI Training Room, DOST Compound, Bicutan, Taguig City	
26	2 PTRI Employees and 2 Contract of Service Members	Solidworks Innovation Day 2019	CADCentric Solutions, Inc./Bellevue Hotel, Alabang, Muntinlupa City	

November 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
6	1 PTRI Employee	"Forum on the Salient Provisions of the Data Privacy Act of 2012"	DOST-National Privacy Commission (NPC)/Platinum Bldg.	

			Auditorium, MIRDC Compound, Bicutan, Taguig City	
8-9	1 PTRI Employee	Orientation-Workshop with the topic entitled "Harmonizing Information Management of the DOST through InfoSerbilis"	STII/Mt. Samat Function Room, 7th floor, BSA Twin Towers Hotel, Ortigas Center, Mandaluyong City	
14-15	2 PTRI Employees	Colloquium on Innovations and Trends in R&D Management	UP/CCDL Auditorium, UP Open University, Los Baños, Laguna	
14-15	1 PTRI Employee	2-day seminar-writeshop on "How to Publish in a Peer-Reviewed Journal"	FNRI/FNRI Training Room, Bicutan, Taguig City	
15	20 PTRI Employees	Orientation/Reorientation Training on Tracking, Retrieval, and Archiving of Communications for Efficiency (TRACE)	PTRI-PICTS/PTRI Training Room	
19	1 PTRI Employee	Seminar on "Awareness on Risk Management (Based on ISO 31000:2009)	MIRDC/MIRDC Compound, Taguig City	P2,500.00
19-23	1 PTRI Employee	"Training Course on Data Visualization Using Infographics (with use of internet)" Batch 3	Philippine Statistical Research and Training Institute (PSRTI), 7th Floor, South Insula Condominium, Timog Ave., Q.C.	
21-24	2 PTRI Employees	4TH Philippine Association for Government Budget Administration (PAGBA), Inc. Quarterly Seminar and Meeting	PAGBA/Grand Menseng Hotel, Davao City	P8,000.00 P8,000.00
26-29	1 PTRI Employee	Assessment of the Internal Control System Training Program	COA/Quezon City	
27-29	3 PTRI Employees	Training on Streamlining and Process Improvement of Critical Services for NGAs	DAP AO Secretariat/Hotel H2O Function Room, Luneta, Manila	
27-28	2 PTRI Employees	Philippine Accreditation Bureau (PAB) Assessor's/Expert's Forum	Philippine Accreditation Bureau (PAB)	
28-29	1 PTRI Employee	2nd National Conference of Chemical Laboratories – "Minimum Standards of International Laboratory Practice: Accuracy and Reliability of Test Results"	Integrated Chemists of the Philippines (ICP)/Century Park Hotel, Manila	P4,500.00 (OneLab)

December 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
3-5	10 PTRI Employees	Training on Computerized Dobby Lim (DOST-GIA Project Perfect Fit)	PTRI/Textile Product Development Center, PTRI	
5	1 PTRI Employee	Yearly OneLab ASEAN Forum	OneLab/Alabang, Muntinlupa City	
5 - 7	1 PTRI Employee	3rd Philippine Solid & Hazardous Waste Management Conference 2018 with the theme: *"The Philippine Waste Outlook: Our Waste, Our Responsibility for a Sustainable Future"*	Bohol Conferences Management Services (BCMS).Bohol Tropics Convention Center, Tagbilaran City, Bohol	Live-in Package of P8,400.00 (inclusive of P6,000 registration fee)
12	1 PTRI Employee	Fourth (4th) Cluster Meeting for CY 2018	CSC-NCR/People's Hall, City Government of Taguig, 9th Floor, SM Aura, Taguig City	
13	2 PTRI Employees	"Appreciation Course on Policy Development and Strategic Planning Processes for the Members of the DOST Management Committee"	DOST/Assembly Hall, Nat'l Defense College of Public Administration and Governance (NCPAG), UP, Diliman, Q.C.	

Planned Trainings/Seminars/Workshops/Activities

August 2018

Duration	Participants		Course/Title	Sponsor/Venue	Fees
9 - 10	86 PTRI Employees including Contract of Service Members		2018 PTRI Team Building (inclusive of partial awarding of PRAISE 2016 and 2017)	PTRI/Bakasyunan Resort and Confere Iba, Zambales	LIB – P367,60
22	23 PTRI Employees including Contract of Service Members		Echo Seminar on Foreign and Local Trainings	PTRI/PTRI Training Room	LIB – P3,860.00

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September 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
6 - 7	77 PTRI Employees including Contract of Service Members	Orientation on the 2017 Rules on Administrative Cases in the Civil Service (2017 RACCS)	CSC-NCR/PTRI Training Room	LIB – P63,524 .00

October 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
16	14 PTRI Employees	Seminar-Workshop on Cardio-Pulmonary Resuscitation (CPR)	Philippine Red Cross/PTRI Training Room	P14,000
19	23 PTRI Employees	Echo-Seminar on Foreign Trainings	PTRI/PTRI Training Room	
24-25	PTRI Employees	Orientation on the Revised 2017 Omnibus Rules on Appointments and Other Human Resource Actions (Revised 2017 ORA OHRA)	CSC-DOST FO/PTRI Training Room	
26	PTRI Employees	"Clean and Green" Program for CY 2018	CSC-DOST FO/PTRI Training Room	

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November 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
14-15	19 PTRI Employees	Livelihood Program for Retirees/Senior Citizens	PTRI-FNRI/PTRI Training Room	

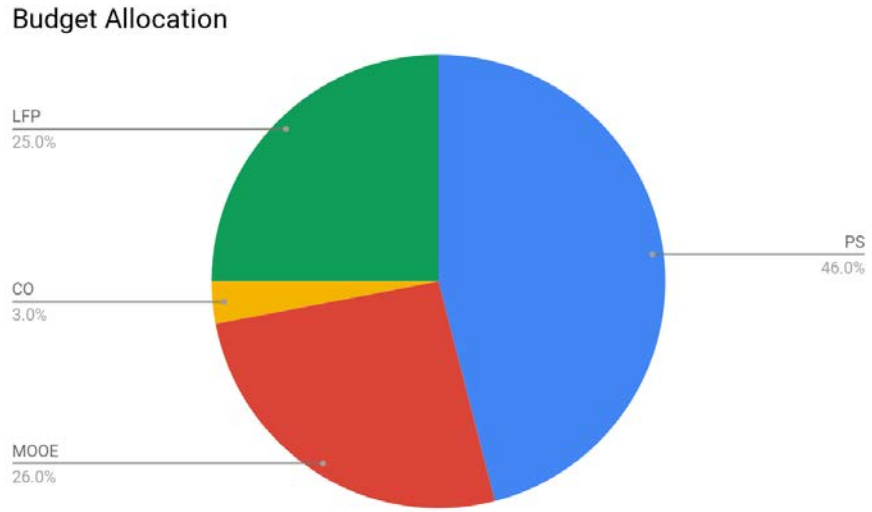
December 2018

Duration	Participants	Course/Title	Sponsor/Venue	Fees
Dec. 20	All PTRI Employees inclusive of Contract of Service	2018 PTRI Year-end General Assembly	PTRI/PTRI Training Room	

[Division 4] **Financial Accountability of PTRI**

The PTRI approved budget as stipulated in the 2018 General Appropriations Act amounted to PHP 121,106,000. Of the total budget, 46% or PHP 55,077,000 was allotted for Personnel Services (PS), 26% or PHP 31,675,000 for Maintenance and Other Operating Expenses (MOOE), 3% or PHP 4,140,000 for Capital Outlay and 25% or PHP 29,903,000 for Locally-Funded Projects.

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Expenditures (in million PHP)

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By Expense Class

Expense Class	Allotment	Obligations	Disbursement
Personnel Services	55,377	55,077	54,746
Maintenance and Other Operating Expenses (MOOE)	31,686	31,675	27,978
Capital Outlay	4,140	4,130	3,972
Locally-Funded Projects	29,903	29,846	21,963
TOTAL	**121,106**	**120,728**	**108,659**

Commented [42]: The budget allotted for a certain project or expenditure

Commented [43]: The amount that the government is obligated to pay as the expenditure for their projects

Commented [44]: The amount that has actually been withdrawn from the government treasury in order to pay for the obligations

By Programs/Projects/Activity (P/A/Ps)

P/A/Ps	Allotment	Obligations[1]	Disbursement[2]
General Administration and Support	41,095	40,849	40,536
Textile and Other Textile-Related Research and Development Program	53,751	53,680	42,359
Textile S&T Services	14,968	14,912	14,625
Textile Technology Transfer Program	11,292	11,287	11,139
TOTAL	**121,106**	**120,728**	**108,659**

[1] *Obligations are liabilities legally incurred and committed to be paid for by the government either immediately or in the future.*
[2] *Disbursements refer to the actual withdrawal of cash from the Bureau of the Treasury due to the encashment of checks issued by agencies and payment of budgetary obligations.*

For the year 2018, the Department of Budget and Management (DBM) released a total cash allotment of PHP 115,487 million to PTRI's Regular Modified Disbursement System (MDS) Account. Of the total cash received under the Regular Fund, a total of PHP 108,659 million was disbursed revealing a 94% utilization rate. Out of PHP 121,106 million allotment released: PHP 120,728 million has been obligated, equivalent to 99.69% of the total allotment. PTRI was able to disburse 108,659 million or 89.72% of its total allotment mentioned above.

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Additional Generated Income

Aside from the PHP 121,106,000 budget provided by the GAA, the PTRI was able to generate an additional PHP 67,333,000 from their completed projects and initiatives.

Funding Source	Project Title	Project Cost (in PHP)
Philippine Commission on Women (PCW)	Support to Revitalization of Philippine Textile and Garment Industry through STI for Women Economic Empowerment	4,880,000
National Historical Commission of the Philippines (NHCP)	Identification of Indigenous Materials and Rock Site as Alternative Compatible Materials in the Restoration of Heritage Structure in the Philippines	1,411,000
National Grid Corporation of the Philippines (NGCP)	An Approach to Improving Competitiveness of Local Weavers of Buhi One Town Product Key Players Association	518,000
DOST-GIA	Commercial Scale Verification of Natural Dyes for Textiles	3,436,000
DOST-GIA	Establishment of Regional Yarn Production and Innovation Center	37,363,000
DOST-GIA	Establishment of PTRI "Perfect Fit" Textile Product Development Center	14,175,000
DOST-GIA	Enhancing OneLab for Global Competitiveness	5,550,000
TOTAL		**PHP 67,333,00**

MOVING FORWARD

[Text pending answers to questionnaire]